THE PONTIFICAL AND ROYAL
UNIVERSITY OF SANTO TOMAS
THE CATHOLIC UNIVERSITY IN THE PHILIPPINES
COLLEGE OF SCIENCE
COLLEGE OF INFORMATION AND COMPUTING SCIENCE

Scalable Machine Learning for Global Internet Performance Tiering Using Ookla Network Speed Data

Bautista, Justine Benedict
Diao, Michael Anthony
Reyes, Darylle Joshua
Tagama, Nathan Lee

Submitted to
Sir Ahdrian Camilo Gernale

In partial fulfillment of the requirements for
Principles of Big Data

May, 2026

TABLE OF CONTENTS

Abstract

Reliable internet connectivity has become a basic requirement for participation in education, commerce, governance, health services, and everyday communication (Milakovich, 2012). However, internet quality remains uneven across regions and connection types, creating a persistent digital divide that cannot be evaluated solely through simple averages and descriptive statistics (Greiman et al., 2023). This study implements a scalable machine learning pipeline to analyze global internet performance using the Ookla Network Speed dataset. The project focuses on implementation rather than purely socioeconomic explanation: it demonstrates how large-scale network performance records can be ingested, cleaned, engineered, transformed, clustered, and modeled using PySpark.

The downloaded dataset, obtained from Kaggle, contained multiple quarterly folders from 2019 to 2021, with both fixed broadband and mobile performance files. The full dataset folder occupied 37.80 GB, while the 24 analysis-ready Parquet performance files used in the machine learning pipeline occupied 10.29 GB. However, the dataset folder was fragmented and contained different file types. The main variables included geographic tile identifiers, average download speed, average upload speed, average latency, test counts, device counts, year, quarter, and connection type. The pipeline converted speed units from kbps to Mbps, created ratio-based variables such as download-to-upload ratio and tests per device, and applied log transformations to reduce skew in speed, latency, test, and device variables.

After performing EDA, three scalable machine learning tasks were implemented. First, K-means clustering was applied to identify three tiers of internet performance. The clustering model produced a silhouette score of 0.2394. Cluster 1 represented the high-performance tier, with the highest average download speed (101.74 Mbps), the highest upload speed (42.10 Mbps), and the lowest average latency (21.26 ms). Cluster 0 represented a stable or high-activity tier, with good speeds and high measurement density. Cluster 2 represented the low-performance tier, with the lowest average download speed (14.86 Mbps), the lowest average upload speed (5.02 Mbps), and the highest average latency (55.69 ms).

Second, Linear Regression was implemented as a baseline model for predicting log-transformed latency, yielding an RMSE of 0.6745, an MAE of 0.4864, and an R^2 of 0.3822. Third, Random Forest Regression was implemented as a stronger non-linear model, yielding an RMSE of 0.6550, an MAE of 0.4696, and an R^2 of 0.4174. Random Forest outperformed the baseline across all metrics, although the improvement was moderate.

Overall, the study shows how important PySpark is when performing scalable machine learning. Hence, the study also shows that scalable machine learning is

capable of organizing large-scale internet speed records into interpretable performance tiers and generating predictive insights about latency. The findings also show that core network variables, especially upload speed, connection type, and download speed, are important predictors of latency. However, the remaining model error suggests that latency is affected by additional factors not fully captured in the dataset, such as routing distance, congestion, server placement, local infrastructure conditions, and other external factors that were not included in the dataset.

Introduction

2.1 Background and Context

Throughout human history, significant periods such as the Agricultural and Industrial Revolutions have fundamentally reshaped societies. Today, this is seen in the Digital Revolution that continues this transformation through rapid advancements in computing, communication, and information technologies. The internet is a direct proponent of this revolution, and can be considered as one of its most significant developments, as it has become critical to how people access information, communicate, learn, and navigate their daily lives.

Reliable internet connectivity has become critical in a number of societal sectors, including education, healthcare, commerce, and governance. The increasing reliance on digital infrastructure highlights the need to address network performance and access.

Despite these technological advancements, a persistent digital divide threatens to limit equitable progress. Identified constraints, such as limited connectivity, insufficient infrastructure, and inconsistent network performance, prevent many individuals from fully participating in digitally enabled sectors. As global internet usage expands, evaluating and addressing these disparities is key to allowing for accessibility and inclusivity.

2.2 Motivation and Connection to Real-World Complexity

This study is motivated by the ongoing digital divide, which results in unequal internet access quality across different regions and populations. Although internet connectivity has expanded globally, network performance remains highly inconsistent due to differences in infrastructure, economic conditions, service availability, and technological development.

The large-scale and dynamic nature of internet performance data introduces significant analytical challenges. Traditional evaluation methods may struggle to efficiently process large datasets and identify meaningful performance patterns across diverse network environments. To address this, scalable machine learning approaches provide a practical framework for analyzing large-scale network speed data and organizing internet performance into meaningful tiers.

Additionally, internet performance datasets often lack predefined classifications that distinguish between high-, medium-, and low-performing network conditions. This creates a need for data-driven approaches that can identify natural groupings in network behavior while maintaining scalability and interpretability.

By applying scalable machine learning techniques to Ookla network speed data, this study aims to provide a structured analysis of global internet performance patterns and demonstrate how large-scale network datasets can be effectively analyzed using distributed machine learning workflows.

Related Work

For studies that look to analyze and address network performance, the Ookla Open Data fits the bill quite well. Its large-scale internet speed dataset, which includes variables such as download speed, upload speed, latency, test counts, and device counts, makes it suitable for studying internet quality across different locations and connection types. Its Parquet format also supports efficient processing for large-scale analysis.

For large datasets such as this one, scalable tools are necessary to analyze them effectively. Tools such as Apache Spark and PySpark enable large Parquet files to be cleaned, transformed, and modeled without loading the entire dataset into local memory (Theodorakopoulos et al., 2025). In this study, PySpark will be utilized to support ingestion, feature engineering, scaling, clustering, and regression modeling.

K-Means clustering was chosen as the appropriate model for this case because the dataset has no predefined labels for high-, moderate-, or low-performing internet conditions (Xiahou & Harada, 2022). It enables the study to identify data-driven tiers of internet performance. For latency prediction, Linear Regression is used as an interpretable baseline, while Random Forest Regression is used to capture more complex non-linear relationships between network variables.

Model & Algorithm

4.1 Theoretical Framework

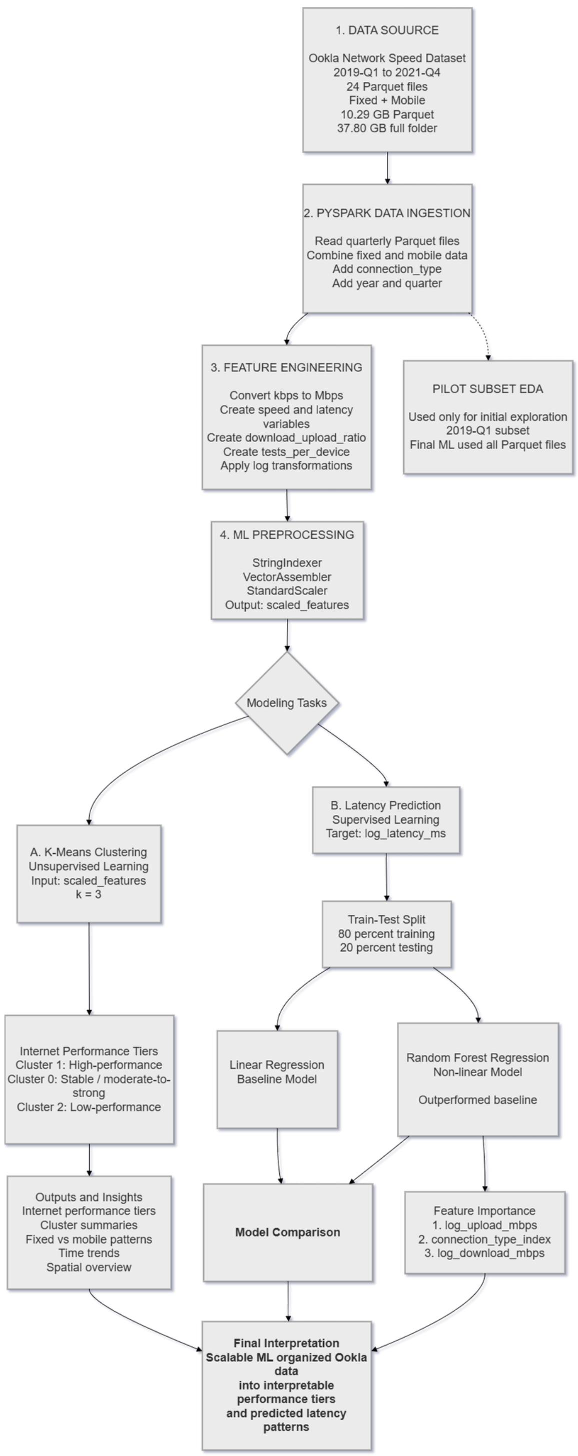

Figure 4.1.1: Pipeline Diagram

4.2 Design Choices

This study uses PySpark for scalable data ingestion, preprocessing, and machine learning, given the large-scale nature of the Ookla network speed dataset. The dataset is stored in Parquet format, enabling efficient storage and faster processing of structured network performance data.

K-Means clustering is selected as the primary machine learning method because the dataset lacks predefined labels for internet performance categories. Its computational efficiency and compatibility with PySpark make it suitable for scalable analysis.

To further analyze network behavior, regression models are also explored for latency prediction. Linear Regression is implemented as a baseline interpretable model, while Random Forest Regression is additionally explored to capture more complex non-linear relationships between network variables. Together, these design choices establish a scalable framework for analyzing and categorizing global internet performance patterns.

Implementation

5.1 Parameters and Conditions

The study utilizes network-related variables from the Ookla internet speed dataset, which contains records of fixed and mobile network performance collected across multiple quarterly Parquet files. The variables include average download and upload speeds, latency, tests, device, year, quarter, and connection type. Additional engineered features, such as download-upload ratio, tests per device, and log-transformed network variables, are generated to support scalable machine learning analysis.

PySpark is used as the primary framework for scalable data ingestion, preprocessing, feature engineering, and machine learning. The implementation utilizes PySpark MLlib within a local Spark environment configured for large-scale data processing. Supporting Python libraries such as pandas, NumPy, and Matplotlib are additionally used for exploratory analysis and visualization on sampled datasets.

For clustering tasks, K-Means is implemented with k = 3, a maximum iteration count of 20, and squared Euclidean distance for silhouette evaluation. Regression analysis uses log-transformed latency as the target variable. Linear Regression is used as the baseline predictive model, and Random Forest Regression is also explored to analyze relationships between network variables and latency behavior.

5.2 Methodology

The pipeline begins by ingesting the dataset into PySpark, loading and combining multiple quarterly Parquet files containing fixed and mobile internet records. Relevant network variables are then cleaned, transformed, and engineered into machine learning features using scalable Spark DataFrame operations.

Feature preprocessing includes categorical encoding, vector assembly, feature scaling, and log transformation of selected network variables. K-Means clustering is subsequently applied to group records into internet performance tiers based on similarities in network behavior. Cluster summaries and distributions are analyzed to characterize global patterns of internet performance.

Regression analysis is additionally performed using Linear Regression to analyze latency behavior and relationships among network variables. Model outputs, evaluation metrics, and visualizations are generated to support the interpretation of clustering behavior and internet performance trends.

Results

6.1 Experimental Findings

The K-Means model was trained successfully on the Spark ML-ready DataFrame using k = 3. The model produced a silhouette score of 0.2394. This indicates that the clusters were moderately separated, although the boundaries are not highly distinct. The value is not high, but this result is reasonable for real-world internet performance data because network quality exists along a continuous spectrum rather than in perfectly separated categories. Therefore, the clusters are interpreted as approximate performance tiers rather than absolute classes.

The counts of records among the clusters turned out to be uneven. Cluster 0 contained 24,345,817, Cluster 1 contained 44,716,375, and Cluster 2 contained 54,295,519. In total, 123,357,711 records were obtained after preprocessing, as shown in the clustering results.

Among all clusters, Cluster 1 shows a high-performance tier as it scores the highest download and upload speeds, and the lowest average latency. Cluster 0 shows a stable or high activity tier performing strong speeds and latency, but much higher average test and device counts. Cluster 2 was interpreted as the low-performance tier because it had the lowest speeds and the highest latency, even if it had the highest records among the clusters.

The connection-type distribution backs up the cluster interpretation. Clusters 0 and 1 were dominated by fixed broadband records, while Cluster 2 contained more mobile records than fixed records. This suggests that the stronger-performing tiers were more associated with fixed connections, while the low-performance tier was more associated with mobile connections.

Over time, Cluster 1 generally increased from 2019 to 2021, while Cluster 2 declined in later quarters. This may suggest a shift toward stronger measured network performance, although changes in test activity, coverage, and device participation may also cause an effect on the record counts of the clusters.

MODEL	RMSE	MAE	R^2
Linear Regression	0.6745	0.4864	0.3822
Random Forest	0.6550	0.4696	0.4174

Table 6.1.1: Model Scores

The model comparison shows that Random Forest Regression outperformed the Linear Regression baseline across all evaluation metrics. Random Forest achieved lower RMSE and MAE, indicating smaller prediction errors on the log-transformed latency target. It also achieved a higher R^2 score, indicating that it explained more variance in latency than the linear baseline.

The improvement suggests that latency prediction benefits from a non-linear model. This is consistent with the earlier Linear Regression plots, where the baseline model compressed predictions into a narrower range and underpredicted records with extremely high latencies. Random Forest was better at capturing some of these complex relationships, though the improvement was moderate rather than dramatic.

Overall, Linear Regression remains useful as an interpretable baseline, while Random Forest provides stronger predictive performance. However, the remaining prediction error suggests that latency is influenced by additional factors not fully captured in the dataset, such as routing distance, congestion, server placement, or local infrastructure conditions.

6.2 Analysis

The K-Means results done in the project show that scalable clustering identified meaningful internet performance tiers. The clusters differed in download speed, upload speed, latency, connection type, and measurement density. Cluster 1 represented the strongest tier, Cluster 0 represented the moderate-to-strong tier, and Cluster 2 represented the weakest tier. The silhouette score of 0.2394 suggests moderate separation, meaning the clusters should be interpreted as approximate performance tiers rather than strict categories, considering their boundaries are not highly distinct.

The regression results show that latency can be partially predicted using the available network variables. Linear Regression explained 38.22% of the variation in log-latency, while Random Forest improved this to 41.74%. Moreover, this suggests that non-linear modeling captured latency behavior better than the linear baseline, although the moderate improvement suggests that other factors, such as routing, congestion, server distance, or infrastructure quality, may also affect latency.

Lastly, the Random Forest feature importance highlights that the strongest predictors of latency are: log_upload_mbps, connection_type_index, and log_download_mbps. Hence, this suggests that latency prediction is mainly associated with core speed variables and whether the connection is fixed or mobile, while measurement-density and time variables contributed less.

6.3 Visualizations

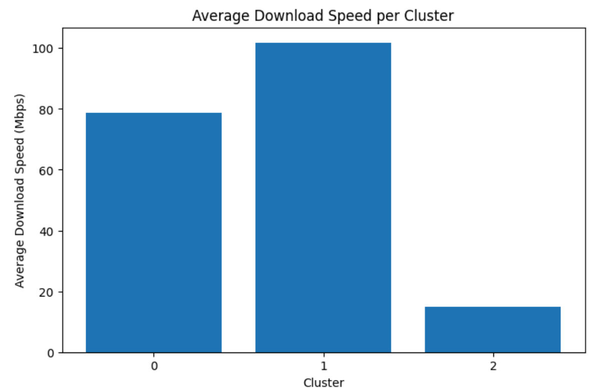

Figure 6.3.1: Average Download Speed

The chart shows clear differences in average download speed across the three K-Means clusters. Cluster 1 has the highest average download speed at around 101.74 Mbps, followed by Cluster 0 at around 78.86 Mbps. Cluster 2 has a much lower average download speed at around 14.86 Mbps.

This suggests that Cluster 1 represents the strongest download-performance tier, while Cluster 0 represents a moderately strong or stable-performance tier. Cluster 2 represents the lowest download-performance tier. The large gap between Cluster 2 and the other two clusters shows that K-Means was able to separate records with substantially weaker download performance from records with stronger network performance.

This result supports the goal of the project because the clustering model produces interpretable internet performance tiers based on observed network behavior.

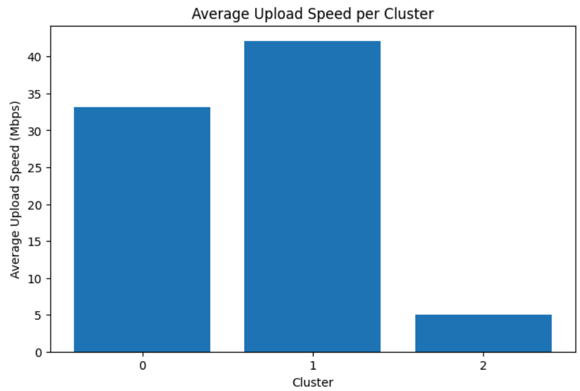

Figure 6.3.2: Average Upload Speed

The upload speed pattern is similar to the download speed pattern. Cluster 1 has the highest average upload speed at around 42.10 Mbps, followed by Cluster 0 at around 33.11 Mbps. Cluster 2 has the lowest average upload speed at around 5.02 Mbps.

This strengthens the interpretation of Cluster 1 as the high-performance internet tier and Cluster 2 as the low-performance tier. The large difference between Cluster 2 and the other clusters suggests that lower-performing records are not only slower in download speed, but also significantly weaker in upload performance.

Upload speed is important because it affects tasks such as video calls, file sharing, online classes, remote work, and cloud-based activities. Therefore, the low upload speed of Cluster 2 indicates a broader limitation in network quality, not just weaker download throughput.

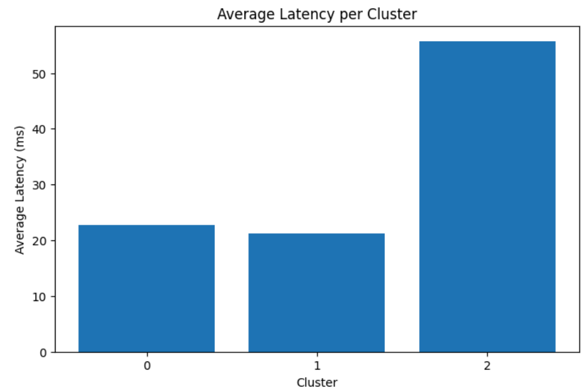

Figure 6.3.3: Average Latency

Since latency was not included in our K-means scaled_features, we're simply using it as a post-clustering validation to confirm and help with interpretations

The latency chart shows that Cluster 2 has the highest average latency at around 55.69 ms, while Cluster 0 and Cluster 1 have much lower average latency values at around 22.75 ms and 21.26 ms, respectively. Since lower latency indicates better network responsiveness, this confirms that Cluster 2 represents the weakest performance tier.

This result strengthens the earlier speed-based interpretation. Cluster 2 is not only slower in download and upload speed, but also less responsive. In contrast, Cluster 1 has the best overall performance because it combines the highest average download and upload speeds with the lowest average latency. Cluster 0 also performs relatively well, although it is slightly slower than Cluster 1.

Overall, the latency pattern supports the interpretation of the clusters as internet performance tiers: Cluster 1 as the high-performance tier, Cluster 0 as the stable/moderate-to-strong tier, and Cluster 2 as the low-performance tier.

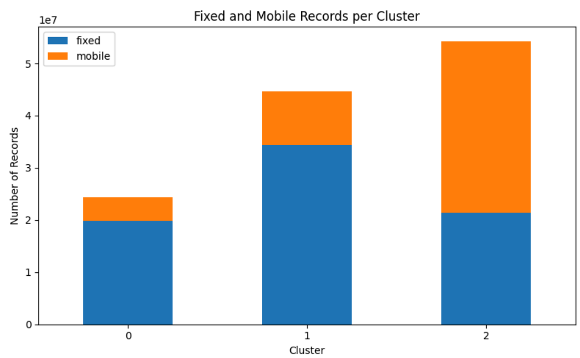

Figure 6.3.4: Fixed and Mobile Records

The stacked bar chart shows the distribution of fixed and mobile records within each K-Means cluster. Cluster 0 and Cluster 1 are mostly composed of fixed connection records, while Cluster 2 contains more mobile records than fixed records.

This pattern supports the earlier performance-tier interpretation. Cluster 1, which had the highest average download and upload speeds and the lowest latency, is strongly dominated by fixed broadband records. Cluster 0 is also mostly fixed and represents a stable or moderate-to-strong performance tier. In contrast, Cluster 2, which had the lowest speeds and highest latency, contains a larger share of mobile records.

This suggests that connection type is an important factor in the clustering structure. Fixed broadband records are more associated with stronger-performing tiers, while mobile records are more concentrated in the lower-performing tier. This may reflect the greater variability of mobile networks compared with fixed broadband infrastructure.

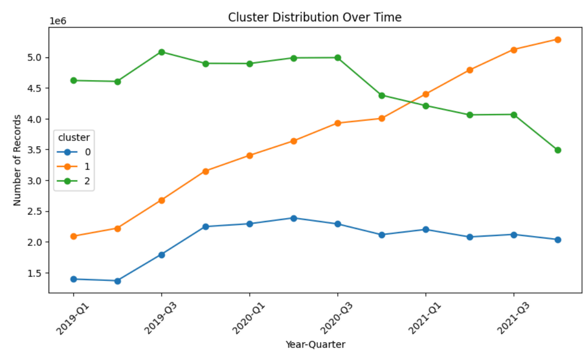

Figure 6.3.5: Cluster Distribution Over Time

The cluster distribution over time shows a noticeable shift in the composition of internet performance records from 2019 to 2021. Cluster 1 steadily increases across the quarters, starting at around 2.1 million records in 2019-Q1 and reaching around 5.3 million records by 2021-Q4. Since Cluster 1 represents the high-performance tier, this suggests that a growing portion of the dataset is being classified into stronger-performing network conditions over time.

Cluster 2 shows the opposite pattern. It starts as the largest cluster in 2019, with more than 4.6 million records, remains high through much of 2020, and then declines sharply in 2021. Since Cluster 2 represents the low-performance tier, this may suggest a reduction in weaker-performing records over time.

Cluster 0 stays relatively stable compared with the other two clusters. It increases from 2019 to early 2020, then remains around 2.0 to 2.4 million records in later quarters. This supports its interpretation as a stable or moderate-to-strong performance tier.

Overall, the time-based distribution suggests a possible shift from lower-performance records toward higher-performance records across the observed period. However, this should be interpreted carefully because changes in record counts may also reflect changes in Speedtest usage, device coverage, data availability, or measurement density rather than infrastructure improvement alone.

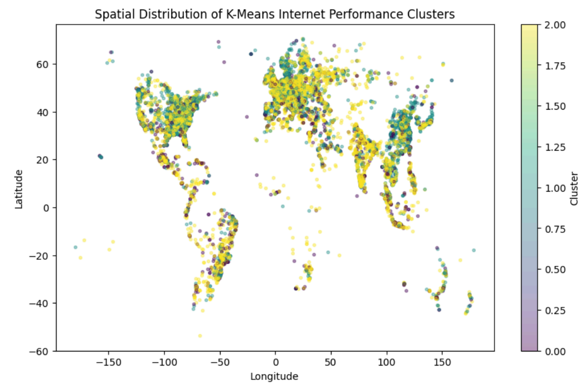

Figure 6.3.6: Spatial Distribution of K-Means Internet Performance

The spatial scatter plot provides a geographic overview of the K-Means clusters using sampled tile centroids. It shows that all three clusters appear across multiple regions, suggesting that the clusters represent global network-performance patterns rather than being limited to specific geographic areas. Cluster 2 appears widely distributed, supporting the finding that weaker network conditions make up a large portion of the dataset. However, since the plot uses sampled records, approximate centroids, and does not include latitude or longitude in the K-Means model, it should be interpreted only as a broad spatial visualization rather than a precise geographic analysis.

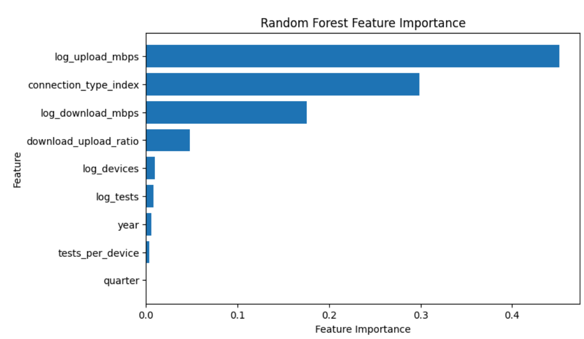

Figure 6.3.7: Random Forest Feature Importance

The Random Forest feature importance plot shows that `log_upload_mbps` is the most influential predictor of log-transformed latency, with an importance value of around 0.4516. This suggests that upload performance contains strong information about network responsiveness. In this dataset, lower or weaker upload performance may be associated with higher latency or poorer network conditions.

The second most important feature is `connection_type_index`, with an importance value of around 0.2988. This indicates that whether a record comes from a fixed or mobile connection is highly relevant for latency prediction. This is consistent with the earlier K-Means results, where fixed and mobile records showed different cluster patterns.

The third most important feature is `log_download_mbps`, with an importance value of around 0.1752. This means download speed also contributes meaningfully to latency prediction, although it is less influential than upload speed and connection type in the Random Forest model.

Other variables, such as `download_upload_ratio`, `log_devices`, `log_tests`, `year`, `tests_per_device`, and `quarter`, have much smaller importance values. This suggests that the model relies more heavily on core network-performance variables and connection type than on time period or measurement density.

Overall, the feature importance results show that latency prediction is mainly driven by upload speed, connection type, and download speed. However, these importance values should not be interpreted as direct causal effects. They only indicate which features the Random Forest model used most when making predictions.

Discussion

7.1 Discussion of Findings & Critical Reflections

Findings from the study reflect that scalable machine learning can identify meaningful patterns in large-scale internet performance data. The K-Means model successfully grouped records into three interpretable performance tiers, representing stronger-performing, moderate-performing, and weaker-performing network conditions. Although the silhouette score of 0.2394 indicates only moderate cluster separation, this is expected because internet performance exists on a continuous spectrum rather than in clearly separated categories.

The regression results further suggest that latency can be partially predicted using available network variables. Random Forest Regression performed better than Linear Regression, indicating that latency behavior is influenced by more complex, non-linear relationships among network conditions. However, the moderate prediction performance also highlights that latency depends on additional real-world factors not captured in the dataset, such as routing distance, network congestion, server location, and infrastructure quality.

A major finding of the project is the importance of scalability in handling large internet performance datasets. PySpark enabled large-scale preprocessing, feature engineering, clustering, and regression analysis without relying entirely on in-memory processing tools such as pandas. This demonstrates how distributed machine learning workflows can support the analysis of real-world network datasets that may otherwise be difficult to process efficiently using traditional approaches.

The findings should also be interpreted carefully. The generated clusters describe patterns in internet performance data, but do not explain the direct causes of network quality differences. Similarly, the regression models identify predictive relationships between variables rather than causal relationships. Despite these limitations, the project demonstrates how scalable machine learning can provide structured and interpretable insights into global internet performance behavior.

7.2 Difficult Aspects

The main difficulty was handling the dataset size. The full downloaded folder was 37.80 GB, while the Parquet files used for machine learning totaled 10.29 GB. Some full-data Spark actions caused runtime and disk space issues; some ran for almost 2 hours. Hence, the workflow had to avoid unnecessary .count(), .cache(), and .toPandas() operations to avoid error and crashing of the VSCode.

Another challenge was balancing scalability and visualization. Full-data plotting was not practical, so pilot samples and aggregated Spark outputs were used for visualizations. K-Means training was also computationally expensive because it is an iterative algorithm, requiring a long runtime (70 minutes on average) even in PySpark.

Finally, interpreting the results required caution and an in-depth understanding of both the dataset and the modeling process. EDA was also challenging because the files were originally fragmented, and the group had to decide how to conduct the EDA to ensure the project ran smoothly. The clusters were meaningful but only moderately separated, and the regression models did not fully explain latency. Therefore, the results were framed as scalable data-driven insights rather than complete explanations of internet performance or the digital divide.

7.3 Limitations

This project has several limitations. First, the exploratory visualization stage used a pilot sample because full-data visualization was not practical on local hardware. However, the main feature engineering, preprocessing, clustering, and regression models were implemented using PySpark on all available Parquet files.

Second, the K-Means silhouette score was 0.2394, indicating that the clusters are only moderately separated. This is expected because real-world internet performance exists on a continuous spectrum rather than in perfectly distinct classes.

Third, the regression models predicted log-transformed latency using only available network-performance variables. Other potentially important factors, such as server distance, network congestion, routing paths, geographic infrastructure, and local service provider conditions, were not included in the dataset.

Finally, the models were implemented in a local Spark environment. Runtime and memory constraints influenced some implementation decisions, especially around visualization and avoiding full-data conversion to pandas.

7.4 Recommendations

Future work would better benefit from incorporating additional geographic and socioeconomic variables. Joining Ookla performance tiles with population density, urbanization, income, infrastructure investment, or electricity access data could improve the interpretation of the digital divide. However, such integration should be carried out carefully to avoid ecological fallacies and mismatches in spatial scales (Gkikas et al., 2024).

Future work should also test additional values of k for K-Means. This project used k = 3 for interpretability, but future experiments could compare k values using the silhouette score, within-cluster sum of squares, and interpretability of resulting tiers.

For latency prediction, future work could implement Gradient-Boosted Tree Regression, tune Random Forest hyperparameters, or compare models using cross-validation (Clark & Lee, 2025). These steps may improve predictive performance, especially by implementing Gradient-Boosted Tree Regression, but they require more computational resources (Ivatt & Evans, 2020).

7.5 Practical Application

There are a number of practical applications that can be applied from the implemented pipeline. It can be useful for organizations interested in monitoring internet performance. Telecommunications researchers could use similar clustering to identify groups of network tiles with similar behavior. Policymakers could use performance tiers to identify areas requiring more detailed investigation. Internet service providers could use similar methods to compare fixed and mobile performance patterns over time.

The approach is also useful for academic big data education. It demonstrates how a large dataset can be processed end-to-end using scalable tools: ingestion, cleaning, feature engineering, unsupervised learning, supervised learning, evaluation, and visualization.

Conclusion

This project implemented a scalable PySpark-based machine learning pipeline for global internet performance tiering using the Ookla network speed dataset. The pipeline processed all available Parquet performance files, engineered network-quality features, applied scalable preprocessing, performed K-Means clustering, and compared supervised regression models for latency prediction.

The K-Means model identified three interpretable performance tiers: a high-performance tier, a stable or moderate-to-strong tier, and a low-performance tier. The cluster summaries showed clear differences in download speed, upload speed, latency, connection type composition, and measurement density.

For latency prediction, Linear Regression served as the baseline model, while Random Forest Regression provided the stronger non-linear model. Random Forest outperformed Linear Regression across RMSE, MAE, and R^2, showing that non-linear modeling better captures latency patterns. However, the improvement was moderate, suggesting that latency is affected by additional factors beyond the available features.

Overall, the project demonstrates how scalable machine learning methods can be used to process and analyze large-scale internet performance data, producing interpretable network performance tiers and predictive insights into latency.

References

Milakovich, M. (2012). Digital Governance. In *Taylor Francis*. https://api.taylorfrancis.com/content/books/mono/download?identifierName=doi&identifierValue=10.4324/9780203815991&type=googlepdf

Greiman, L., Leopold, A., Ipsen, C., & Myers, A. (2023). Mapping the digital divide: What predicts internet access across America? *Local Development & Society*, 1–13. https://doi.org/10.1080/26883597.2023.2229962

Theodorakopoulos, L., Karras, A., & Krimpas, G. A. (2025). Optimizing Apache Spark MLlib: Predictive Performance of Large-Scale Models for Big Data Analytics. *Algorithms*, *18*(2), 74. https://doi.org/10.3390/a18020074

Xiahou, X., & Harada, Y. (2022). B2C E-Commerce Customer Churn Prediction Based on K-Means and SVM. *Journal of Theoretical and Applied Electronic Commerce Research*, *17*(2), 458–475. https://doi.org/10.3390/jtaer17020024

Gkikas, M. C., Gkikas, D. C., Gerasimos Vonitsanos, Theodorou, J. A., & Spyros Sioutas. (2024). Application of Machine Learning for Predictive Analysis and Management of Mediterranean-Farmed Fish Mortalities: A Risk Management Case Study Using Apache Spark. *Applied Sciences*, *14*(22), 10112–10112. https://doi.org/10.3390/app142210112

Clark, B., & Lee, F. (2025, April 7). *What is Gradient Boosting?* Ibm.com. https://www.ibm.com/think/topics/gradient-boosting

Ivatt, P. D., & Evans, M. J. (2020). Improving the prediction of an atmospheric chemistry transport model using gradient-boosted regression trees. *Atmospheric Chemistry and Physics*, *20*(13), 8063–8082. https://doi.org/10.5194/acp-20-8063-2020

Dave, D. (2020). *Internet Speed Dataset*. Kaggle.com. https://www.kaggle.com/datasets/dhruvildave/ookla-internet-speed-dataset?fbclid=IwY2xjawR9VHdleHRuA2FlbQIxMQBzcnRjBmFwcF9pZAEwAAEeNj58lOrPAKfoS6bZE0pkRFgGTj74uqzZ04J4LUQOUtOH5CsyGc3fXsOdkug_aem_NCb9jP7zTtg7NrfdaFDROQ

AI Disclosure:

Artificial Intelligence was utilized during the drafting process to assist with grammatical corrections, sentence restructuring, stylistic refinements, and brainstorming model selection. All final content was reviewed and approved by the authors.

GitHub Link:

https://github.com/justinebenedictb/Big-Data-Final-Project